

Differentially Private Algorithms for Linear Queries via Stochastic Convex Optimization

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Query release problem

1. **Record space** $\mathcal{Z} = \{\mathbf{z}_1, \dots, \mathbf{z}_k\}$, $\mathcal{P}$ supported on $\mathcal{Z}$.
2. **Private Dataset** $\mathcal{D}_n \stackrel{i.i.d}{\sim} \mathcal{P}^n$ sampled from $\mathcal{Z}$, $\mathcal{D}_n = \{\mathbf{z}_{k_1}, \dots, \mathbf{z}_{k_n}\}$
3. **Query Class** $\mathcal{Q} = \{\mathbf{q}_1, \dots, \mathbf{q}_{|\mathcal{Q}|}, \mathbf{q} : \mathcal{Z} \rightarrow [-1, 1]\}$
4. **Statistical Query** $\mathbb{E}_{\mathbf{z} \sim \mathcal{P}}[\mathbf{q}(\mathbf{z})]$
5. **Simplex** $\Delta_k = \{\mathbf{p} \in [0, 1]^k : \sum_i \mathbf{p}_i = 1\}$
6. **Objective**: Find a private distribution $\mathcal{P}^{\text{priv}}$ on $\mathcal{Z}$ such that the error in answering all queries is minimized

$$\min_{\mathcal{P}^{\text{priv}} \in \Delta_k} \max_{\mathbf{q} \in \mathcal{Q}} |\mathbb{E}_{\mathbf{z} \sim \mathcal{P}}[\mathbf{q}(\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim \mathcal{P}^{\text{priv}}}[\mathbf{q}(\mathbf{z})]|$$

subject to: $\mathcal{P}^{\text{priv}} = \mathcal{A}(\mathcal{D}_n)$
 $\mathcal{A}$ satisfies (ϵ, δ) -DP (P)

7. **Meaning**: we privatize $\mathcal{P}$ by releasing $\mathcal{P}^{\text{priv}}$ instead of answering $\mathbb{E}_{\mathbf{z} \sim \mathcal{P}}[\mathbf{q}(\mathbf{z})]$, and report the private estimate $\mathbb{E}_{\mathbf{z} \sim \mathcal{P}^{\text{priv}}}[\mathbf{q}(\mathbf{z})]$.

ID	Start Date	Age	Salary
1	02/01/2010	35	60.000
2	08/15/2012	43	54.000
3	11/03/2009	36	62.000
4	09/12/2013	42	56.000
5	04/05/2008	24	69.000
6	06/28/2014	40	58.000
7	12/17/2011	54	75.000

Ex. $\mathbf{q}(\mathbf{z}) = \mathbf{1}\{\mathbf{z} \text{ is older than } 38\}$
 $\rightarrow \mathbf{z}_1 \rightarrow \mathbf{q}(\mathbf{z}_1)$

$\vdots$
 $\vdots$

$\rightarrow \mathbf{z}_7 \rightarrow \mathbf{q}(\mathbf{z}_7)$

- **Non-private query (percentage > 38):**

$$\frac{1}{7} \sum_{i=1}^7 \mathbf{q}(\mathbf{z}_i) = \mathbb{E}_{\mathbf{z} \sim \mathcal{P}_n}[\mathbf{q}(\mathbf{z})] \quad (\text{non-private empirical mean})$$

- **Private query (percentage > 38):**

$$\mathbb{E}_{\mathbf{z} \sim \mathcal{P}^{\text{priv}}}[\mathbf{q}(\mathbf{z})] \quad (\text{based on private distribution } \mathcal{P}^{\text{priv}})$$

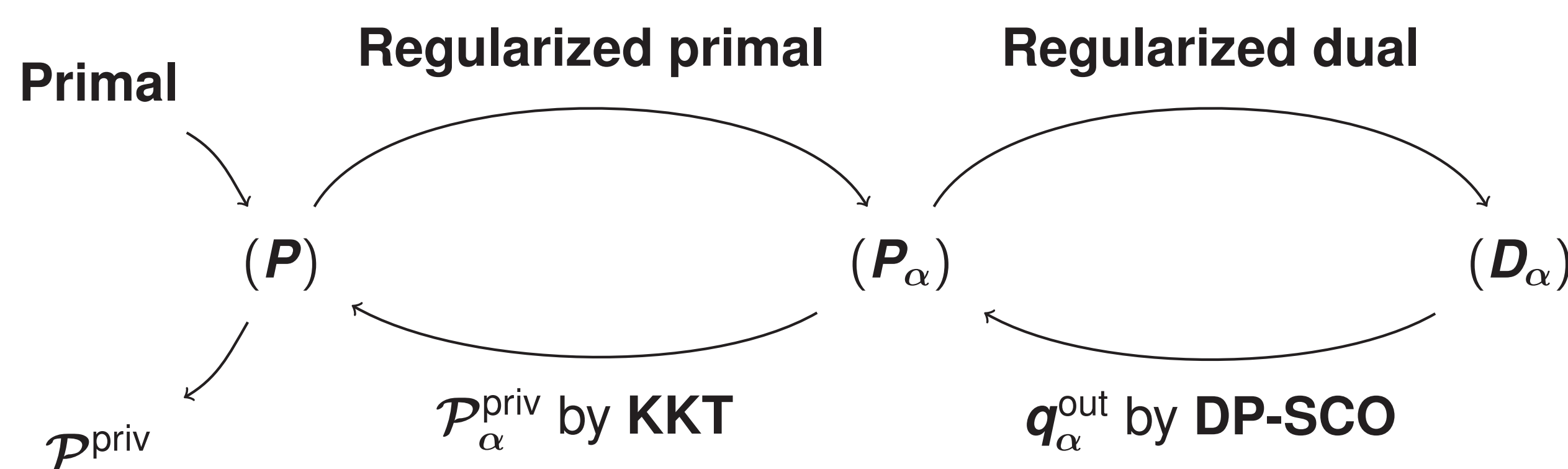
Regularization I

$$\min_{\mathcal{P}^{\text{priv}} \in \Delta_k} \max_{\mathbf{q} \in \mathcal{Q}} \mathbb{E}_{\mathbf{z} \sim \mathcal{P}}[\mathbf{q}(\mathbf{z})] - \mathbb{E}_{\mathbf{z} \sim \mathcal{P}^{\text{priv}}}[\mathbf{q}(\mathbf{z})] + \alpha H(\mathcal{P}^{\text{priv}}) \quad (\mathbf{P}_\alpha)$$

$$\max_{\mathbf{q} \in \text{conv}(\mathcal{Q})} \langle \mathbf{q}, \mathcal{P} \rangle - \alpha H^*\left(\frac{\mathbf{q}}{\alpha}\right) \quad (\mathbf{D}_\alpha)$$

where H is the negative entropy and $H^*(\mathbf{y}) = \log\left(\sum_j e^{y_j}\right)$ is the Fenchel conjugate of H .

Advantage: $(\mathbf{D}_\alpha)$ is strongly concave $\rightarrow$ solve via DP-SCO.



Primal-Dual Gap

$$\mathbb{E}[\text{Gap}_{(\mathbf{P}_\alpha)}(\mathcal{P}_\alpha^{\text{priv}})] \leq \mathbb{E}[\text{Gap}_{(\mathbf{P}_\alpha)}(\mathbf{q}_\alpha^{\text{out}})] + \alpha \log k$$

Optimal α is chosen to minimize the right-hand side.

DP-Frank Wolfe

Differentially private Frank-Wolfe algorithm (DPFW)

- 1: **Input:** $(\mathcal{D}_n, \mathcal{Q}, (\epsilon, \delta), \alpha)$
- 2: Initialization: $\mathbf{q}_0 \in \mathcal{Q}$, T num. it and γ step size
- 3: **for** $t = 0, \dots, T - 1$ **do**
- 4: Compute $\hat{\nabla} \psi_\alpha(\mathbf{q}_t)$ (gradient estimator for objective $(\mathbf{D}_\alpha)$)
- 5: For all $\mathbf{s} \in \mathcal{Q}$ sample $\mathbf{u}_{s,t} \stackrel{i.i.d}{\sim} \text{Lap}(\lambda(\epsilon, \delta))$
- 6: $\mathbf{s}_t = \text{argmax}_{\mathbf{s} \in \mathcal{Q}} \{\langle \hat{\nabla} \psi_\alpha(\mathbf{q}_t), \mathbf{s} \rangle + \mathbf{u}_{s,t}\}$
- 7: $\mathbf{q}_{t+1} = \mathbf{q}_t + \gamma(\mathbf{s}_t - \mathbf{q}_t)$
- 8: **Output:** $\mathbf{q}_\alpha^{\text{out}} = \mathbf{q}_U$ for $U \sim \text{Uni}(\{0, \dots, T - 1\})$

Error Bounds from Dual to Primal

DP-Frank Wolfe produces $\mathbf{q}_\alpha^{\text{out}}$ that is (ϵ, δ) -DP and satisfies:

$$\mathbb{E}[\mathbf{g}_{\psi_\alpha}(\mathbf{q}_\alpha^{\text{out}})] = \mathcal{O}\left(\frac{\log^{1/4}(1/\delta) \log^{1/2}(|\mathcal{Q}|)}{\alpha^{1/4}(\epsilon n)^{1/2}} + \sqrt{\frac{\log(k)}{n}}\right)$$

Optimizing over α yields α^* , and setting $\mathcal{P}^{\text{priv}} = \mathcal{P}_{\alpha^*}^{\text{priv}}$ leads to:

$$\mathbb{E}[\text{Gap}_{(\mathbf{P})}(\mathcal{P}^{\text{priv}})] = \mathcal{O}\left(\frac{\log^{1/5}(1/\delta) \log^{2/5}|\mathcal{Q}| \log^{1/5}k}{(\epsilon n)^{2/5}} + D_1 \sqrt{\frac{\log k}{n}}\right)$$

Regularization II: Smoothed Objective

$$\min_{\mathcal{P}^{\text{priv}} \in \Delta_k} \left(\mathbb{E}_{\xi \sim \mathcal{N}} \left[\max_{\mathbf{q} \in \mathcal{Q}} \langle \mathbf{q}, \mathcal{P} - \mathcal{P}^{\text{priv}} + \xi \rangle \right] + \alpha H(\mathcal{P}^{\text{priv}}) \right) \quad (\mathbf{P}_{\alpha, \sigma})$$

Privacy is injected directly into the objective via **randomized smoothing**. $\mathcal{N} := \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$

DP-Accelerated Mirror Descent

Differentially Private Complementary Accelerated Mirror-Descent

- 1: **Input:** $(\mathcal{D}_n, \mathcal{Q}, (\epsilon, \delta), \alpha)$
- 2: Initialization: $T = T(\epsilon, \delta)$ num. it = $\sigma(\epsilon, \delta)$, Initialisation $\mathcal{P}_1 = \mathcal{P}_1^{\text{ag}} = [1/k, \dots, 1/k]^\top$
- 3: Step sizes: $\eta_t = t + \sqrt{\frac{4}{\alpha \sigma}} + 1$, for $1 \leq t \leq T$
- 4: **for** $1 \leq t \leq T$ **do**
- 5: $\mathcal{P}_t^{\text{md}} = \frac{\sum_{\tau=1}^{t-1} \eta_\tau \mathcal{P}_\tau^{\text{ag}}}{\sum_{\tau=1}^{t-1} \eta_\tau} + \frac{\eta_t}{\sum_{\tau=1}^t \eta_\tau} \mathcal{P}_t$
- 6: $\mathbf{g}_t = \mathbf{G}_\sigma(\mathcal{P}_t^{\text{md}}, \xi_t, \mathcal{D}_n, \mathcal{Q})$, $\xi_t \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_k)$ (gradient estimator for objective $(\mathbf{P}_{\alpha, \sigma})$)
- 7: $\mathcal{P}_{t+1} = \text{argmin}_{\mathcal{P} \in \Delta_k} \left\{ \eta_t [\langle \mathbf{g}_t, \mathcal{P} \rangle + H(\mathcal{P})] + \left(\sum_{\tau=1}^{t-1} \eta_\tau \right) D_H(\mathcal{P}, \mathcal{P}_t) \right\}$
- 8: $\mathcal{P}_{t+1}^{\text{ag}} = \frac{\sum_{\tau=1}^{t-1} \eta_\tau \mathcal{P}_\tau^{\text{ag}}}{\sum_{\tau=1}^{t-1} \eta_\tau} + \frac{\eta_t}{\sum_{\tau=1}^t \eta_\tau} \mathcal{P}_{t+1}$
- 9: **Output:** $\mathcal{P}_\alpha^{\text{priv}} = \mathcal{P}_{T+1}^{\text{ag}}$.

Primal Gap via Smoothed Objective

Algorithm 2 produces $\mathcal{P}_\alpha^{\text{priv}}$ that is (ϵ, δ) -DP and satisfies:

$$\mathbb{E}[\text{Gap}_{(\mathbf{P}_{\alpha, \sigma})}(\mathcal{P}_\alpha^{\text{priv}})] = \mathcal{O}\left(\frac{\log^{3/4}(1/\delta)}{\alpha \log^{5/4}(k)} \cdot \frac{w^{5/2}(\mathcal{Q})}{(n\epsilon)^{3/2}} + \frac{\log^{1/2}(1/\delta)}{\alpha \log^{1/2}(k)} \cdot \frac{w(\mathcal{Q})}{n\epsilon}\right)$$

Optimizing over α yields α^* such that:

$$\mathbb{E}[\text{Gap}_{(\mathbf{P}_{\alpha^*, \sigma})}(\mathcal{P}_{\alpha^*}^{\text{priv}})] \leq \mathcal{O}\left(\frac{w^{1/2}(\mathcal{Q}) \log^{1/4}(k) \log^{1/4}(1/\delta)}{\epsilon^{1/2} n^{1/2}}\right)$$

where $w(\mathcal{Q})$ is the **Gaussian width** of the query class $\mathcal{Q}$.

Established	Empirical Loss
MWEM	$\mathcal{O}\left(\frac{d^{1/4} \log^{1/2} \mathcal{Q} \log^{1/2}(1/\delta)}{n^{1/2} \epsilon^{1/2}}\right)$
DualQuery	$\mathcal{O}\left(\frac{d^{1/6} \log^{1/2} \mathcal{Q} \log^{1/6}(1/\delta)}{n^{1/3} \epsilon^{1/3}}\right)$
FEM	$\mathcal{O}\left(\frac{d^{3/4} \log^{1/2} \mathcal{Q} \log^{1/2}(1/\delta)}{n^{1/2} \epsilon^{1/2}}\right)$
sepFEM	$\mathcal{O}\left(\frac{d^{5/8} \log^{1/2} \mathcal{Q} \log^{1/2}(1/\delta)}{n^{1/2} \epsilon^{1/2}}\right)$
DQRS	$\mathcal{O}\left(\frac{d^{1/5} \log^{3/5} \mathcal{Q} \log^{1/5}(1/\delta)}{n^{2/5} \epsilon^{2/5}}\right)$
DPFW(E)	$\mathcal{O}\left(\frac{d^{1/5} \log^{2/5} \mathcal{Q} \log^{1/5}(1/\delta)}{n^{2/5} \epsilon^{2/5}}\right)$
New	Population Loss
DPFW(P)	$\mathcal{O}\left(\frac{d^{1/5} \log^{2/5} \mathcal{Q} \log^{1/5}(1/\delta)}{n^{2/5} \epsilon^{2/5}} + D_1 \frac{\log^{1/2}(d)}{n^{1/2}}\right)$
DPAM	$\mathcal{O}\left(\frac{d^{1/4} w^{1/2}(\mathcal{Q}) \log^{1/4}(1/\delta)}{n^{1/2} \epsilon^{1/2}}\right)$

